

Lecture 4: Brownian Motion & Random Lattice Walks

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Based on lecture notes by me and many previous CS 335 instructors

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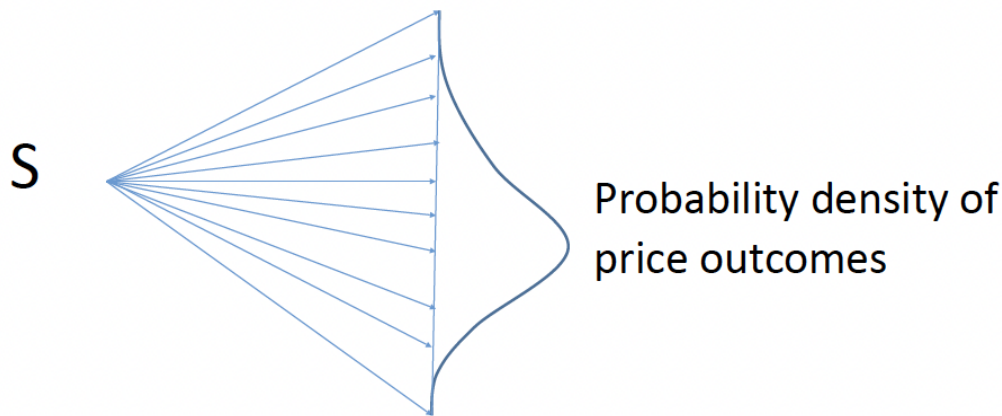
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Option Pricing

The two-state tree model illustrates some essential concepts of option pricing. In particular ...

- We do not attempt to predict the underlying asset price movements.
- Rather we assume that risky assets (e.g., stocks) follow a “stochastic process”, i.e., they are random and unpredictable.

The two-state model is limited, however: real stock prices have a *probability distribution* of potential outcomes instead of 2 choices:



Modeling Uncertainty

To model the uncertainty/randomness of stock price movements, we will use Brownian motion and stochastic differential equations.

Let $X(t)$ be a random, time-dependent variable. During a small time period during which $t \rightarrow t + dt$, we assume that $X \rightarrow X + dX$, where dX obeys:

$$\frac{dX}{dt} = \alpha \quad \leftarrow \quad dX = \alpha dt + \sigma dZ$$

- αdt is the drift term (deterministic / non-random).
- σ is the volatility (a measure of how much noise there is).
- $dZ = \phi \sqrt{dt}$ is a random term, with ϕ being a random number from the standard normal distribution, i.e., $\phi \sim N(0, 1)$ (It is also referred to as the increment of a Wiener process.)

Modeling Uncertainty

I have provided you with a notebook to build an intuitive understanding of the behaviour of our model. Try changing α , dt , σ and the number of simulations! Even try playing around with different distribution settings or types of distributions:

<https://colab.research.google.com/drive/19JqLVKPtZN1NOHjXEdqdjszL1exrz6Vn?usp=sharing>

- What do you think $E[dX]$ is? $E[X]$?
- What do you think $var[dX]$ is?
- How many simulations does it take to start noticing predictable behaviour?

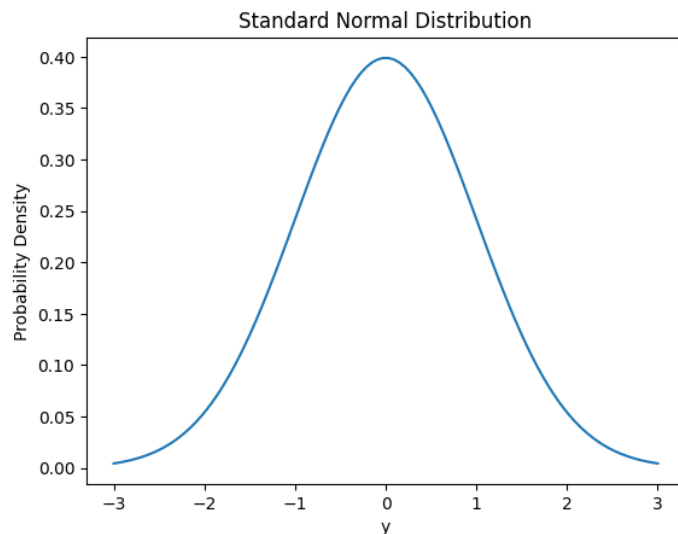
Refresher: Normal Distributions

The probability density function (PDF) for a standard normal distribution, denoted $N(0, 1)$, is:

$$p(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}}$$

The probability of y drawn from such a distribution falling within a given range is expressed as:

$$\text{prob}(a \leq y \leq b) = \int_a^b p(y) dy$$



General Normal Distributions

More generally, for $y \sim N(\mu, \beta^2)$ the density is given by:

$$p(y) = \frac{1}{\sqrt{2\pi\beta^2}} e^{-\frac{(y-\mu)^2}{2\beta^2}}$$

with

$$E[y] = \frac{\text{mean / Expectation}}{\text{of } y} = \int_{-\infty}^{+\infty} y p(y) dy = \mu$$

$$\text{var}[y] = \frac{\text{Variance}}{\text{of } y} = E[(y - E[y])^2] = E[y^2] - (E[y])^2 = \beta^2$$

So for $y \sim \mathcal{N}(0, 1)$ we have $E[y] = 0$, $\text{var}[y] = 1$.

Brownian Motion

For our Brownian motion (with drift), the random term was:

$$dZ = \phi \sqrt{dt} \quad \text{with} \quad \phi \sim \mathcal{N}(0, 1)$$

Hence, we know the mean/expectation of ϕ satisfies:

$$E[\phi] = 0$$

and variance

$$\begin{aligned} \text{var}[\phi] &= 1 = E[\phi^2] - \underbrace{(E[\phi])^2}_0 \\ &= E[\phi^2] \end{aligned}$$

Brownian Motion - Mean/Expectation

Now, consider the statistical behaviour of the total movement of the variable X over time, described by:

$$dX = \alpha dt + \sigma dZ$$

Its expectation is $E[dX] = E[\alpha dt + \sigma dZ] = E[\alpha dt] + E[\sigma dZ]$

The first term is deterministic, so $E[\alpha dt] = \alpha dt$

For the second term, $E[\sigma dZ] = \sigma E[dZ] = \sigma E[\phi \sqrt{dt}] = 0$

since σ and dt are deterministic and the expectation of the normal distribution ϕ is 0.

Thus we have $E[dX] = \alpha dt$

Brownian Motion: Variance

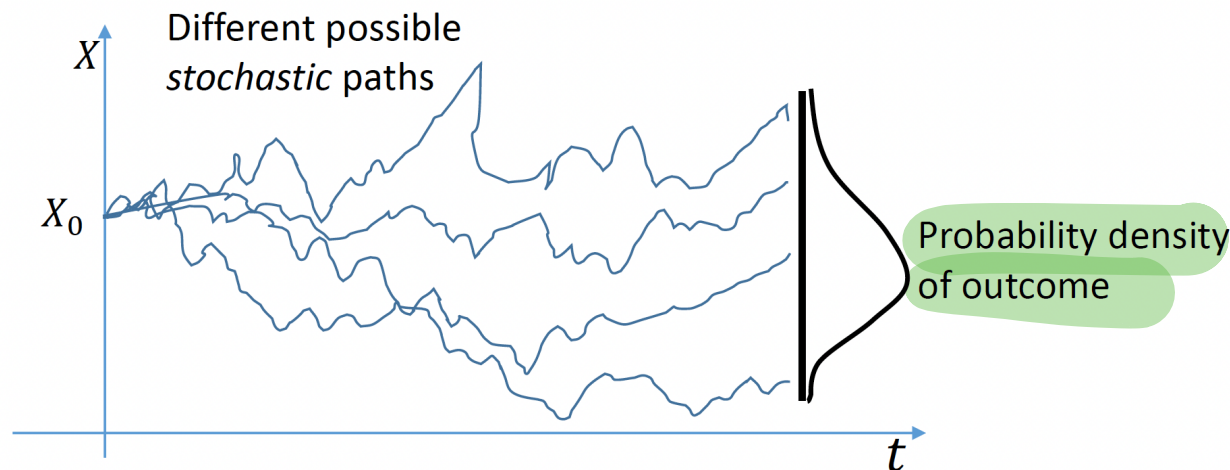
The variance of dX can similarly be worked out, by applying the definitions on the preceding few slides:

$$\begin{aligned} \text{var}[dX] &= E[(dx - E[dx])^2] \\ &= E[(dx - \alpha dt)^2] \\ &= E[(\sigma dz)^2] \\ &= \sigma^2 E[(\phi \sqrt{dt})^2] \\ &= \sigma^2 dt E[\phi^2] \\ &= \sigma^2 dt \underbrace{= 1} \end{aligned}$$

Stochastic Differential Equations

$dX = \alpha dt + \sigma dZ$ (Brownian motion with drift) is an example of a stochastic differential equation (SDE) due to the presence of the random (stochastic) term dZ .

There is no **single solution** to an SDE, only a probability density of outcomes!



A Different View of Brownian Motion

We would like to gain further intuition for Brownian motion, and also find a way to approximate it *numerically* (i.e., *computationally*).

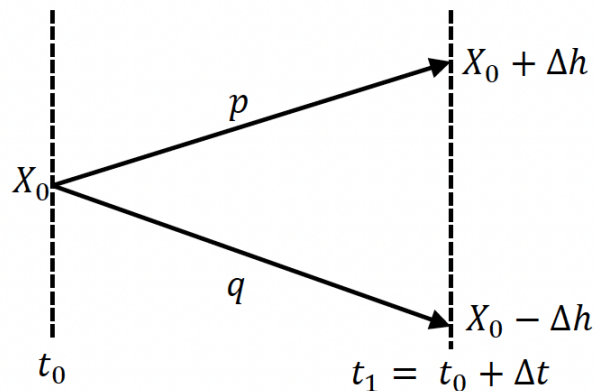
We will consider a discrete *lattice model*, where at each time step, X can (randomly) move either up or down by a fixed amount.

We'll call this a random walk.

A Different View of Brownian Motion

Let $X(t_i)$ be the “position” (i.e., value) of a “particle” (variable) following Brownian motion with drift, at time instant t_i .

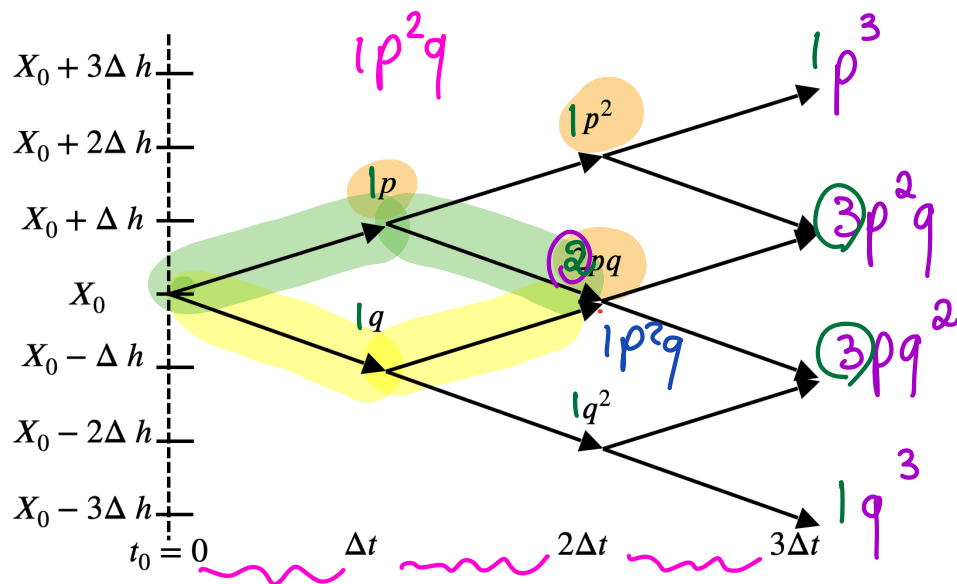
At $t_0 = 0$, $X(t_0) = X_0$, If we step forward to $t_1 = t_0 + \Delta t$, we have:



where $p + q = 1$, and p, q indicate probabilities of an “up” or “down” step, of size Δh (respectively).

Building the Lattice

If we repeated this for several time steps, we get a lattice structure of possible values, at each step forward in time. The probability of arriving at each lattice point is dictated by the history of steps.



End of Lecture 4

Probabilities sum to 1 at each time level. (Notice Pascal's triangle emerges in the factors!)

Lattice Model - Assumptions

We assume X follows a _____:

- Probability of an up/down move is independent of past moves.
- Thus, the probability distribution of future value depends only on where we are now (current value).

Some notation:

- $X_i = X(t_i)$
- $\Delta X_i = X_{i+1} - X_i$

Let's now think about the outcomes (in terms of probability distributions) resulting from such a random walk process.

Expectation of ΔX_i (one step)

ΔX_i are independent and identically distributed (i.i.d.).

$\Delta X_i, \Delta X_j$ are pairwise independent (for $i \neq j$).

i.e., implying $E[\Delta X_i \Delta X_j] = E[\Delta X_i] E[\Delta X_j]$

A single step yields a change by ΔX_i as:

So,

$$E[\Delta X_i] =$$

$$E[\Delta X_i^2] =$$

Variance of ΔX_i (one step)

We can then work out the variance as well:

$$\begin{aligned}\text{var}[\Delta X_i] &= E[(\Delta X_i)^2] - (E[\Delta X_i])^2 \\&= (\Delta h)^2 - (p - q)^2(\Delta h)^2 \\&= (1 - (p - q)^2)(\Delta h)^2 \\&= (1 - (p - (1 - p))^2)(\Delta h)^2 \\&= (1 - (2p - 1)^2)(\Delta h)^2 \\&= (1 - (4p^2 - 4p + 1))(\Delta h)^2 \\&= (-4p^2 + 4p)(\Delta h)^2 \\&= (\Delta h)^2 4p(1 - p) \\&= (\Delta h)^2 4pq\end{aligned}$$